

Edge Computing Systems: A Performance Analysis for Real-Time IoT Data Processing

Lee Yin Shen
Faculty of Computing
Universiti Teknologi Malaysia
Johor, Malaysia
leeshen@graduate.utm.my

Abstract—The rapid growth of connected systems and data-intensive applications has exposed the limitations of traditional centralized computing models, particularly in latency-sensitive environments. Edge computing has emerged as a distributed computing approach that processes data closer to its source, enabling faster response times and improved bandwidth efficiency. This paper examines the architecture and functional characteristics of edge computing systems, including their hierarchical structure, key features, and performance behaviour. A comparative analysis with centralized computing models highlights the advantages of edge computing in reducing latency and supporting real-time processing, while also acknowledging its limitations in resource capacity and system complexity. A real-world case study in automotive manufacturing demonstrates the effectiveness of edge computing in enhancing operational efficiency through localized data processing. Furthermore, key challenges such as resource constraints, security risks, and management complexity are discussed along with potential solutions, including hybrid edge-cloud architectures and AI-based optimization. The findings indicate that edge computing is a critical enabler for real-time applications, offering a balanced approach to distributed data processing.

Index Terms—Edge computing, Internet of Things (IoT), real-time processing, distributed computing, low latency

I. INTRODUCTION

The rapid growth of connected systems and the Internet of Things (IoT) has led to a significant increase in digital data, with more than 75 billion devices expected to be in operation by 2025 [2]. Traditional centralized cloud computing models face critical limitations in this landscape, including high latency, massive bandwidth consumption, and network congestion. Edge computing has emerged as a distributed computing solution to address these challenges by positioning computational resources closer to data sources [2], [6]. Maintaining real-time processing capabilities is essential for time-sensitive applications such as autonomous vehicle navigation, industrial safety systems, and remote healthcare monitoring [2]. This paper aims to analyze edge computing architectures and evaluate their performance characteristics in delay-sensitive and real-time data processing environments. The rest of this paper explains the technological foundations of edge computing, provides a comparative analysis with traditional computing models, and discusses real-world case studies to illustrate the impact of localized processing [2].

II. BACKGROUND AND RELATED WORK

Traditional cloud computing has long served as the primary paradigm for managing distributed applications and large-scale data storage [1]. However, studies have shown that its centralized architecture introduces performance limitations, particularly in latency-sensitive environments, due to data transmission delays and network dependency [1], [2], [4]. These constraints reduce the efficiency of cloud systems when handling real-time workloads that require immediate responses. To overcome these challenges, edge computing has been proposed as a distributed computing approach that shifts processing closer to data sources. This paradigm enables faster response times and supports delay-sensitive applications, especially within Internet of Things (IoT) environments such as autonomous vehicles, industrial systems, and healthcare monitoring [2], [4].

III. TECHNOLOGY OVERVIEW

A. Overview of Edge Computing System

Edge computing represents a distributed computing model that migrates network, storage, and computational capabilities from centralized infrastructures to the edge of the network [3]. The basic principle involves proximity-based processing, where resources are placed as close as possible to data generators to enable localized execution [2]. Conceptually, this model operates by utilizing a diverse array of nearby devices, such as gateways and micro-data centers, to provide immediate intelligent services. Unlike centralized computing, which concentrates all analytical tasks in distant repositories, edge systems redistribute these responsibilities across the network topology [2]. This architectural change allows preliminary data filtering and analysis to occur at the point of origin before relevant insights are transmitted for global integration.

B. Edge Computing Architecture

The architecture of edge computing systems normally follows a three-layer hierarchical structure, including device, edge, and cloud layers, as shown in Fig. 1 [4]. At the basic layer, the device layer consists of spatially distributed sensors and actuators that collect raw environmental data. The intermediate edge layer serves as the core processing tier, utilizing local nodes such as base stations and gateways to perform localized computation and storage [2]. Finally,

the cloud layer provides high-performance infrastructure for global data integration and complex analytics. Data flows sequentially through these layers: raw information is generated at the device level, undergoes preliminary analysis at the edge layer, and actionable insights are subsequently transmitted to the cloud for long-term preservation [2].

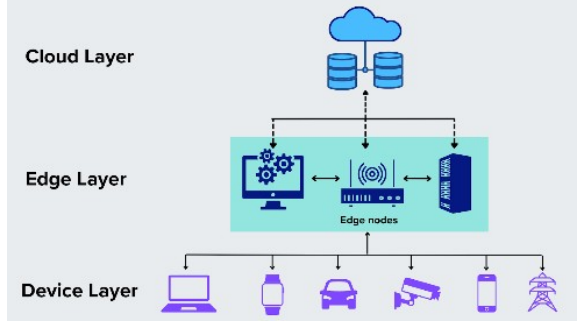


Fig. 1. Edge Computing Layers

C. Key Features of Edge Computing

Edge computing systems have several functional features, mainly including ultra-low latency and location awareness facilitated by physical proximity to end users [5]. Bandwidth optimization is achieved through selective data transmission, where only filtered results are sent across the network, significantly reducing traffic loads. Another key characteristic is operational flexibility, which allows edge nodes to maintain essential processing functions independently during cloud connectivity interruption [2]. Furthermore, these systems support mobility management that enables applications to switch between distributed resources as users move through different geographical regions. In short, these features provide a framework for situational awareness services that require deterministic feedback and high reliability to ensure the continuity of functions in dynamic environments, because centralized resources may be inaccessible or inefficient.

D. Performance Characteristics

Performance in edge systems is characterized by a significant improvement in response times. Empirical data shows that delay is reduced by up to 95% compared with the centralized model [2]. These systems operate with high efficiency by significantly lowering network loads, often reducing traffic by 30% to 90% through localized processing [2]. This behavior is critical for maintaining the deterministic performance required by delay-sensitive applications such as industrial control systems and autonomous navigation. However, performance behavior is governed by the inherent resource constraints of edge nodes, which possess limited computational power, memory capacity, and energy reserves compared to cloud centers [2]. Therefore, complex tasks may experience performance regression, and queuing delays at high utilization levels can offset the physical proximity advantage of the edge [1].

IV. COMPARATIVE ANALYSIS: EDGE COMPUTING VS CENTRALIZED COMPUTING

Comparing edge computing with traditional centralized models is important to identify the optimal architectural alignment for various data processing requirements. Centralized cloud computing uses powerful remote clusters for comprehensive data analysis and global integration, while edge computing redistributes intelligence to the network periphery to enable proximity-based processing [3], [7]. This architectural difference significantly improves real-time capability by reducing the high network latency inherent in long-distance data centers [1]. In addition, local execution optimizes bandwidth efficiency through selective data transmission, which effectively reduces network congestion compared with the transmission of large volumes of data in centralized models. While edge systems provide the necessary response time for latency-sensitive applications, centralized infrastructure still has advantages in large-scale processing, long-term archiving, and complex global analysis [2], [3]. Therefore, the choice between these two models depends on the balance between the need for instant local feedback and the need for high-capacity computing depth.

TABLE I
COMPARISON BETWEEN CENTRALIZED CLOUD COMPUTING AND EDGE COMPUTING

Criteria	Centralized Cloud Computing	Edge Computing
Processing Location	Centralized processing in remote data centers	Distributed processing near data sources
Latency	High due to long data transmission	Lower due to proximity-based processing
Bandwidth Usage	High due to continuous data transfer	Reduced through selective data transmission
Real-Time Capability	Limited for time-sensitive applications	Strong support for real-time processing
Data Processing Scale	Suitable for large-scale and complex analysis	Limited by local computational resources
Data Storage	Supports long-term storage and global integration	Focuses on local and temporary data handling
Best Use Case	Large-scale analytics and data-intensive tasks	Delay-sensitive and real-time applications

V. CASE STUDY: EDGE COMPUTING IN REAL WORLD APPLICATION

In automotive manufacturing, edge computing is deployed to enhance real-time quality control through localized image analysis [2]. A prominent implementation involves integrating hundreds of high-definition cameras throughout a production line to detect defects in components. Instead of transmitting approximately two terabytes of daily inspection data to a distant cloud, which previously introduced latencies of up to 20 seconds, edge servers perform computer vision algorithms in situ [2]. This architectural shift enables defect detection in less than 300 milliseconds, facilitating immediate intervention and decreasing defective units by 37% [2]. Furthermore, this approach reduces network bandwidth requirements by 85% because only identified anomalies are transmitted to central systems for global logging [2]. Such significant performance

gains illustrate how edge computing provides the deterministic response times essential for delay-sensitive industrial environments, where even minor processing lags can result in substantial operational costs and production line stoppages, thus improving overall system reliability [2]–[4].

VI. CHALLENGES AND SOLUTIONS

The implementation of edge computing requires solving inherent resource constraints, as the limited computing power and energy of edge nodes restrict the complexity of processing tasks. The hybrid edge–cloud architecture addresses this limitation by allocating intensive workloads to the cloud while maintaining time-sensitive processing at the edge [2], [6]. In addition, the distributed characteristics of edge systems introduce security risks, including increased attack surfaces and potential physical tampering [2], [3]. These risks can be reduced through security mechanisms such as trusted execution environments and secure virtualization technologies [6]. Due to the existence of heterogeneous devices and diversified communication protocols, management complexity also arises. This challenge can be solved by automated orchestration tools and software-defined networks, which can achieve centralized control and efficient resource management [4]. Finally, scalability in a distributed environment can be improved by resource optimization based on artificial intelligence, which can dynamically adjust the system configuration according to workload needs [4].

VII. CONCLUSION

This study explores the architecture and functional characteristics of edge computing systems in the modern data environment. By moving data processing closer to the data source, edge computing can achieve lower latency performance and higher bandwidth efficiency. Comparative analysis shows that edge computing provides powerful real-time responsiveness for latency-sensitive applications, effectively reducing the latency and congestion problems associated with centralized systems. However, its performance is constrained by resource limitations and the complexity of distributed node management. To solve these challenges, future research should focus on hybrid edge–cloud architecture, enhanced security mechanisms, and resource optimization based on artificial intelligence. In general, edge computing has become a key driver for real-time applications by balancing tasks between local and centralized systems.

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